Sprint Report

PORTFOLIO TASK 2/3/4

Unit code: EAT40005
Unit Name: Engineering Project A
Submission date: 19/04/2026

Student Name	Student Id	Statement of contribution to the report
Rumeth Gunasinghe	104498227	Sprint Plan, Sprint Progress, Sprint Demonstration sections were done by Rumeth
Thinidu Madurapperuma	104675729	Sprint Review, Retrospect, Lessons learned sections were done by Thinidu

ACKNOWLEDGMENT OF COUNTRY

We acknowledge the Traditional Owners of the lands on which this work was completed. We recognise the Wurundjeri Woi Wurrung people of the Kulin Nation as the custodians of the land in Melbourne, Australia, where this project work was undertaken. We pay our respects to their Elders past, present, and emerging.

We acknowledge their continuing connection to land, waters, and culture, and recognise the important role that Indigenous knowledge and perspectives play in shaping sustainable and respectful technological development.

As team members working and studying in Australia, we recognise and respect the cultural heritage of the Traditional Owners of the lands on which we live and learn.

1. ESPRINT PLAN

The project is structured into three progressive sprints, each delivering a key stage of development aligned with the overall goal of implementing predictive control for humanoid systems. Sprint 1 focuses on building a strong foundation by studying and implementing both PID and Model Predictive Control (MPC) on a simplified inverted pendulum system. The deliverables for this sprint include a working simulation environment, successful implementation of both controllers, and a comparative analysis of their performance in terms of stability, response, and robustness. This phase establishes core understanding of control strategies and system dynamics.

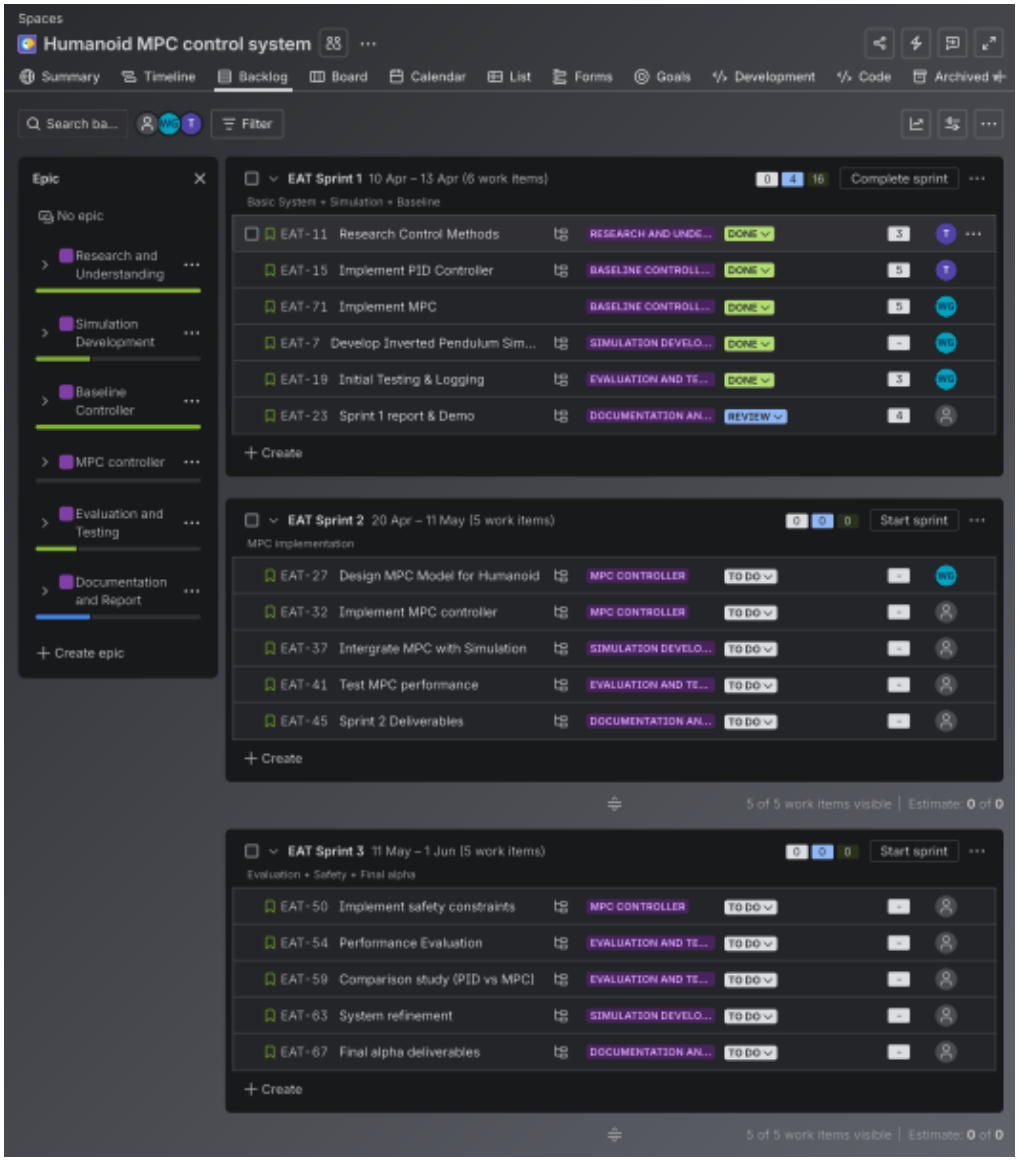


Figure 1 : Sprint plan with estimated tasks allocations per sprint in Jira : https://student-team-cfbljmif.atlassian.net/jira/software/projects/EAT/boards/69?atlOrigin=eyJpIjoiNzE0N2I5YzM2N2FlNDcxYjk2M2MwNTU1MjdjOWJjZWMiLCJwIjoiaiJ9 :

Sprint 2 extends this foundation toward the actual project domain by applying MPC concepts to a humanoid-like system. This involves adapting the control logic from the inverted pendulum to a more complex multi-body structure, and integrating it within a simulation environment such as PyBullet or a similar physics engine. The key deliverables include a functional humanoid simulation model, initial implementation of MPC for maintaining balance or simple motion, and successful integration between control and simulation layers.

Sprint 3 focuses on refining the system with safety and performance considerations. This includes incorporating constraints into the MPC framework (e.g., joint limits, stability constraints, safe interaction conditions) and conducting systematic testing under various scenarios, including disturbances. The deliverables for this phase include a constraint-aware control system, evaluation metrics (such as stability, recovery time, and safety compliance), and a final performance analysis comparing different approaches. This phase ensures the system is not only functional but also reliable and aligned with real-world industrial requirements.

2. SPRINT PROGRESS

Sprint 1 focused on building the core foundation of the project by implementing and evaluating control strategies for an inverted pendulum system. The sprint goal was to develop a working simulation and implement both PID and Model Predictive Control (MPC), followed by initial testing and comparison. This goal was successfully achieved through collaborative contributions from both team members.

From a development perspective, Rumeth Gunasingha was primarily responsible for implementing the MPC controller, designing the predictive control logic, and integrating it with the physics-based simulation. This included developing the cost function, tuning parameters such as prediction horizon and force limits, and ensuring the controller worked with realistic cart-pole dynamics. Additionally, Rumeth contributed to the inverted pendulum simulation development and handled testing scenarios, including disturbance-based evaluations and performance logging.

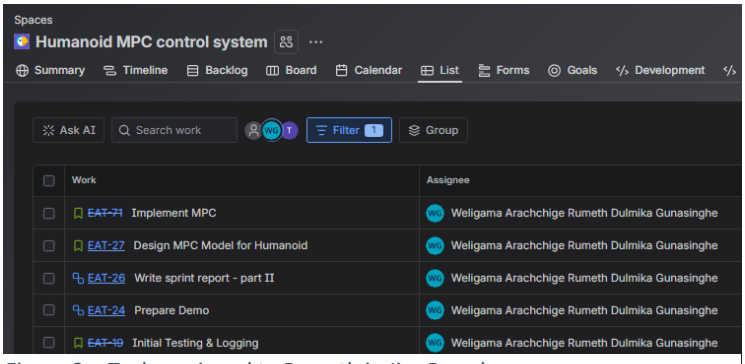


Figure 2: : Tasks assigned to Rumeth in Jira Board

Thinidu focused on implementing the PID controller, including tuning proportional and derivative gains to achieve stable balancing. He also contributed to testing and logging system performance, ensuring that both controllers could be compared under consistent conditions. In

addition, Thinidu worked on documentation and report writing, including drafting sections of the sprint report and maintaining structured documentation of the system.

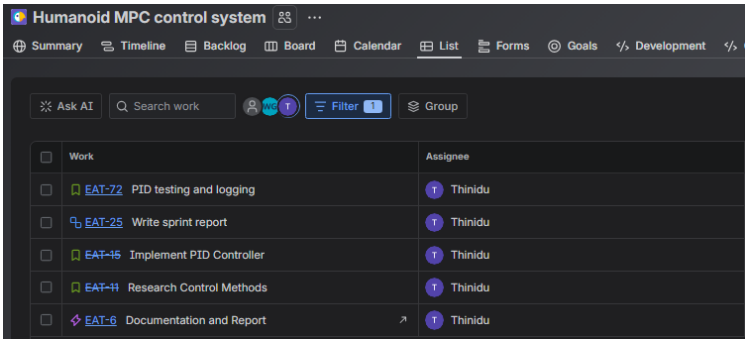


Figure 3: Tasks assigned to THinidu In Jira Board

The sprint deliverables include:

3. A fully functional inverted pendulum simulation using realistic dynamics
4. Implementation of both PID and MPC controllers
5. Performance data collection (angle, position, force graphs)
6. Initial comparison of controller behaviour and stability
7. Structured documentation and sprint reporting

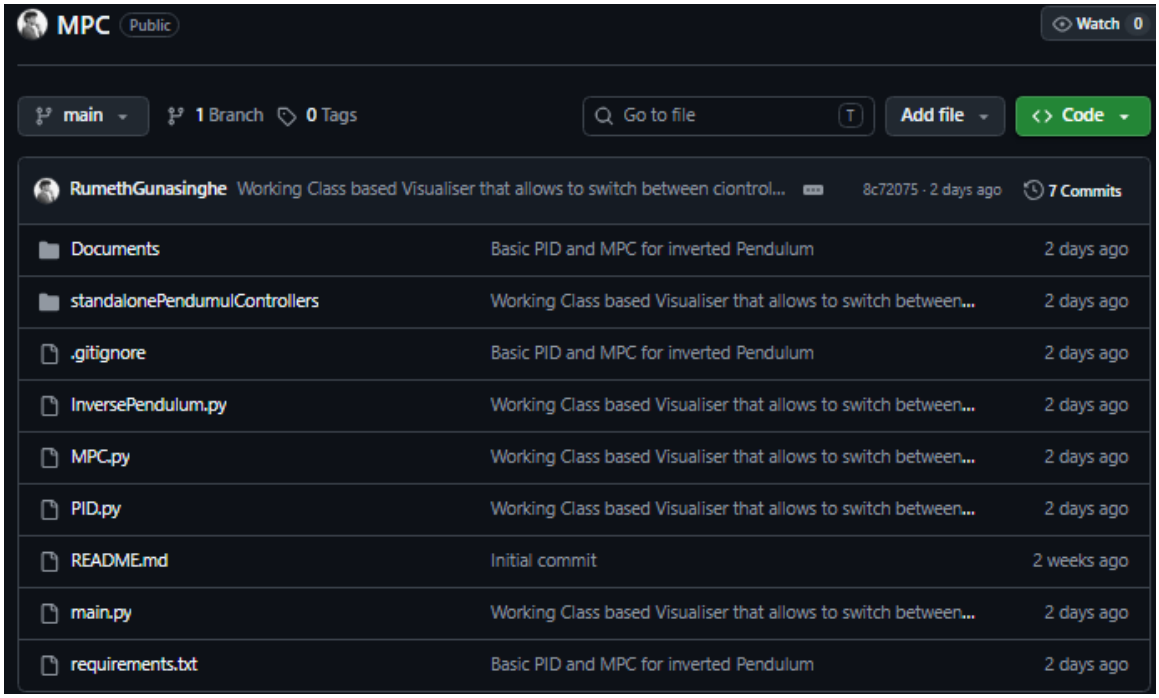


Figure 4: Current Project repo with latest updated files pushed by Thinidu and Rumeth : https://github.com/RumethGunasinghe/MPC.git

8. SPRINT DEMONSTRATION

The sprint demonstration session was conducted on 17/04/2026 with the participation of both team membersand the client representative. The meeting was held via teams.

The objective of this session was to present the team's initial understanding of the project domain and validate the direction of the proposed solution. The team demonstrated their research on humanoid robot stability, focusing on the challenges of dynamic balance during manipulation tasks. Additionally, an early simulation of the cart-pole system was presented to illustrate fundamental control concepts relevant to humanoid balance.

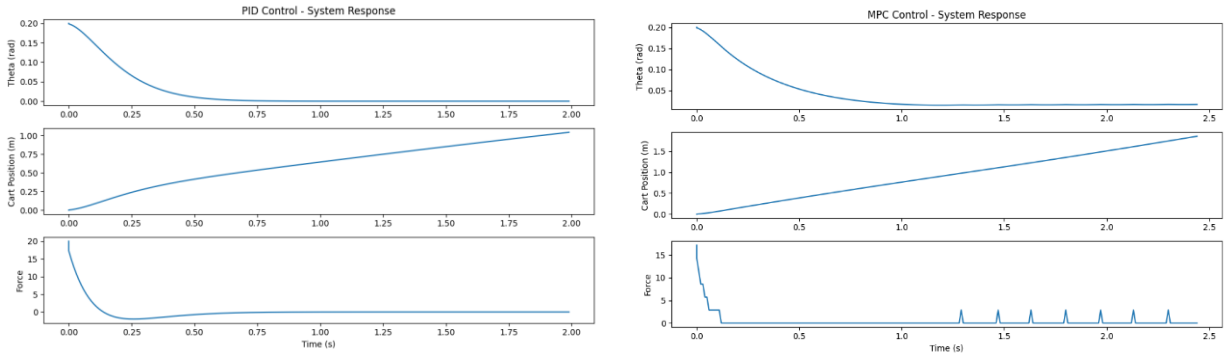


Figure1: PID and MPC controller outputs in a Inverse Pendulum

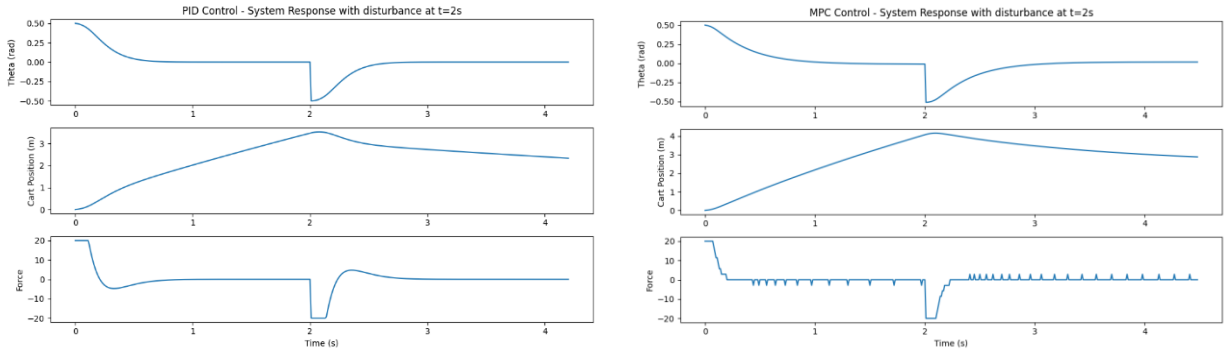


Figure 2: Figure 1: PID and MPC controller outputs in a Inverse Pendulum with Disturbances (0.5rad) added

During the demonstration, the team explained the difference between traditional control methods such as PID and advanced approaches such as Model Predictive Control (MPC). The simulation results showed that both controllers were capable of stabilising the system, while highlighting differences in behaviour and adaptability.

The client provided feedback on the importance of incorporating safety considerations and emphasised the relevance of predictive control in human-robot interaction scenarios. Further clarification was provided regarding the project scope, particularly the expectation to focus on simulation-based validation rather than physical implementation at this stage.

The team also asked several questions regarding expected deliverables, preferred tools, and the level of complexity required for the final system. Based on the discussion, it was agreed that the project would prioritise simulation, control design, and analysis of system behaviour under disturbances.

As an outcome of the session, the project direction was confirmed, and the team will proceed with refining the control models, improving system stability, and incorporating safety constraints in future iterations. The next sprint will focus on enhancing the predictive control implementation and conducting comparative analysis between different control strategies.

9. SPRINT REVIEW (CRITICAL REVIEW OF THE PRODUCT)

The progress of the sprint was evaluated against the initial plan, which aimed to develop a foundational understanding of the problem domain and implement an initial simulation for testing control strategies. The planned deliverables included research on humanoid robot stability, exploration of control techniques, and the development of a basic cart-pole simulation to represent balancing behaviour.

Overall, the team was able to achieve most of the planned objectives. The simulation environment was successfully implemented, and both PID and Model Predictive Control (MPC) approaches were developed and tested. The system demonstrated the ability to balance the pole, providing a strong foundation for further development. This indicates that the sprint goal was achieved.

A key achievement of this sprint was the successful implementation of two different control strategies and the ability to observe their behaviour under simulation. This allowed the team to gain practical insights into how control systems operate in dynamic environments. Additionally, the team developed a clear understanding of the relationship between cart movement and pole stability, which is directly relevant to the broader humanoid robotics problem.

However, several challenges were encountered during the sprint. One of the main challenges was tuning the control parameters to achieve stable behaviour. In particular, balancing the trade-off between stability and drift proved difficult, as overly aggressive constraints caused instability, while weaker constraints led to continuous movement. Another challenge was understanding the behaviour of the MPC controller, especially in relation to prediction horizon and cost function design. These required iterative testing and adjustments.

Feedback from the client during the demonstration emphasised the importance of focusing on safety and reliability in control systems, particularly in human-robot interaction scenarios. The client also highlighted the need for a clear justification of the chosen control approach and its relevance to real-world applications. This feedback influenced the team to place greater emphasis on analysing system behaviour under disturbances and ensuring that the control system responds appropriately to unexpected conditions.

From a reflective perspective, the sprint highlighted the importance of incremental development and testing. Rather than attempting to implement complex features immediately, breaking the problem into smaller components proved more effective. The team also recognised the need for better planning in terms of parameter tuning and experimentation, as this process required more time than initially expected.

For future sprints, the team plans to refine the MPC implementation, improve disturbance handling, and conduct a structured comparison between control methods. Additionally, more attention will be given to documenting results and linking technical outcomes to real-world applications in humanoid robotics.

10. RETROSPECT (CRITICAL REVIEW OF THE PROCESS)

During Sprint 1, the team follows the planned agile process by doing meetings, dividing responsibilities, tracking progress, and maintaining communication from shared channels. As the project scope was initially unclear and detailed client requirements was delayed, the team had to continue using assumptions based on the provided topic: predictive control for industrial humanoids with focus on safe reinforcement learning and stability. This made uncertainty in early stage planning, which required multiple revisions to backlog items once more information became available.

A key challenge was project starting delay, as the team was only finalized after the first few weeks. This narrowed the available sprint time and compressed planning, research, and implementation of tasks into a restricted period. To get over from this, the team prioritized foundational tasks such as literature review, environment setup, and basic prototype development.

Communication was also another challenge due to coordinating schedules and making sure all members remained aligned. Regular online discussions and progress updates helped us to reduce misunderstandings. also, clearer deadlines and more regular communications would have improved the visibility of individual progress.

From a quality perspective, early outputs focused were more on functionality other than structured validation. When simulations for PID and MPC control were developed successfully, more formal testing metrics such as settling time, overshoot, disturbance recovery, and comparison benchmarks should have been defined earlier.

About the risk management, the team identified risks such as unclear requirements, time constraints, unequal workload contribution, and technical difficulty in implementing advanced control systems. These were slightly mitigated through task sharing, continuous communication, and focusing on a simple cart pole model before going to humanoid systems.

Ethically, the team considered that humanoid robots operating in industrial environments must prioritize human safety, predictable behavior, and fail safe control. This strengthens the

importance of researching safe reinforcement learning and formal safety guarantees other than focusing only on performance.

Overall, Sprint 1 established a solid technical foundation, but stronger planning discipline, clearer communication cycles, and earlier quality benchmarks would improve future sprints

11. LESSONS LEARNED (CRITICAL REVIEW OF SPRINT ONE EXPERIENCE AND FUTURE PLAN)

Lessons Learned

- Clear sub scope each sprint from the client is crucial for accurate sprint planning and realistic task estimation.
- Early stage prototyping helped the team understand how the system behave faster than theory alone.
- Regular communication is important to avoid duplicated work and delays.
- Simplifying the problem first (cart pole model) was effective before executing the full humanoid control.
- Quality checks needs to be integrated continuously not than left to the end of the sprint.
- Risk registers have to be actively reviewed and updated, not just create once.
- Safety and ethical implications are essential in robotics projects where systems may interact with humans.

To-Do for Next Sprint

1. Doing short weekly meetings with clear action items and deadlines.
2. Use Jira more actively for task ownership, sprint tracking, and accountability
3. Begin extending from cart pole simulation toward multi link humanoid balancing models.
4. Continuing literature review on safe reinforcement learning and formal safety methods.
5. Reassess team workload assigning to ensure balanced contribution from all members.
6. Maintain professional meeting preparation with agendas, demonstrations, and documented outcomes.
7. Updating risk and communication plans at the beginning of each sprint.